Mini-Challenge 2
Audio-Reactive Spectrum Visualiser



The challenge

Your challenge is to develop an audio-reactive spectrum visualiser using the components provided in your project box.

The system should receive an audio or electrical input, analyse its frequency content and represent that information meaningfully on an LED matrix. The display should respond in real time as the input changes.

Existing prototypes, code and development notes are available to help you. Your task is not simply to reproduce one of them. You should investigate how the system works, decide what a successful product should do, test its behaviour and develop a response that your team can explain and justify.

What are you trying to achieve?

Over the six sessions, your team should aim to develop and demonstrate a system that:

- responds meaningfully to changes in the frequency content of an input signal;
- presents the resulting spectrum clearly on the LED display;
- operates over a frequency range that your team can define and justify;
- behaves predictably under the conditions for which it was designed;
- brings together signal acquisition, processing, visualisation and user interaction; and
- reflects engineering decisions supported by testing and evidence.

A successful project is not necessarily the most complicated one. A well-understood system with clearly established capabilities and limitations is more valuable than sophisticated code that your team cannot explain, test or modify.

The engineering challenge

A spectrum visualiser brings together several stages of an engineering system:

Input signal → signal conditioning → sampling → frequency-domain processing → spectrum mapping → visual display

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Choices made at any stage may affect the final result. A display that looks convincing is not necessarily an accurate representation of the input.

Previous prototypes have demonstrated that the concept can work, but they have also exposed practical challenges involving sampling, noise, filtering, sensitivity, frequency mapping, display resolution and the capabilities of the ESP32. These limitations are part of the challenge rather than faults to be hidden.

Your team should decide which aspects of the system require the greatest attention and what constitutes acceptable performance.

Working through the CDIO cycle

The project will take you through the broad stages of **Conceive, Design, Implement and Operate**.

Conceive

Explore the existing resources and the components in your project box. Decide what your spectrum visualiser is intended to do, who might use it and how you will judge whether it is working successfully.

Design

Develop an approach to the complete system. Consider how the signal will be acquired, processed and represented, which limitations matter, and how you will obtain evidence to support your decisions.

Implement

Build, adapt, integrate and test your chosen solution. You may use existing code and circuits as starting points, but you should understand what you use and make purposeful changes rather than treating the system as a black box.

Operate and Improve

Use the system under a range of conditions. Compare its behaviour with your expectations, identify limitations or unexpected effects, and decide which improvements would create the greatest value.

These stages will overlap. Testing may reveal a poor design assumption, implementation may change your understanding of the problem, and an apparent improvement may introduce a new compromise.

Use the six sessions to develop your response to the challenge. Your team is responsible for deciding how to organise the work, divide responsibilities and use the available time.

Freedom to explore

You are encouraged to:

- experiment with the components and existing software;
- use controlled test signals as well as real audio;
- investigate alternative approaches to sampling, processing and display;
- adapt existing code where it provides a useful starting point;
- compare predicted and observed behaviour;
- pursue an improvement that your team considers worthwhile;
- learn from approaches that do not work as expected; and
- seek advice from demonstrators when you need it.

Possible areas for exploration include input methods, frequency range, sampling behaviour, filtering, noise reduction, amplitude scaling, spectrum mapping, smoothing, selectable display ranges, user controls and physical presentation. These are possibilities, not a list of required features.

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If you use existing code, circuits or designs, acknowledge the source and ensure that your team understands, tests and adapts what it uses.

Questions to guide your thinking

As the project develops, consider questions such as:

- What should the display communicate to the user?
- What does “accurate enough” mean for this product?
- Over what frequency range does the system behave reliably?
- How will you establish whether a displayed peak corresponds to the input frequency?
- How do sampling rate, processing method and display resolution affect what can be shown?
- Could the system produce a plausible but misleading display?
- What happens close to or beyond the useful operating range?
- How does the system behave during silence, rapid changes or complex audio?
- What trade-offs are you making between accuracy, responsiveness, stability and visual appeal?
- How might a user know that the system is operating correctly?
- Could the visual output be difficult for some users to interpret or perceive?
- If a microphone is used, what privacy or ethical considerations arise?
- If wireless features are introduced, are there any security implications?
- Could the system be made easier to repair, adapt, reuse or operate efficiently?
- What evidence would convince another engineer that your conclusions are justified?
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You will not necessarily address every question. Select those most relevant to your design and the stage your project has reached.

Resources and starting points

The following resources show how earlier students developed and tested the spectrum visualiser. They contain useful ideas, working approaches, limitations and possible improvements. They are sources of evidence and inspiration, not model answers.

- Aidan’s Spectrum Visualiser Project Blog
  Development and testing of an ESP32 and MAX7219-based visualiser, including sampling, FFT processing, display mapping, selectable frequency ranges and controlled testing.

- Casey’s Lab-in-a-Box Project Blog
  Analysis of earlier prototypes, identified limitations, mitigation strategies and considerations for turning the system into an inquiry-led laboratory activity.

- Spectrum Visualiser GitHub Repository
  Code and project files from the earlier prototype. Use these as a starting point to understand, test and adapt, rather than as a finished solution.

- Audio/Music Spectrum Visualizer Using Arduino and Dot Matrix Display
  One of the original sources of inspiration for the project.

- 32-Band LED Audio Spectrum Analyser
  A further example of spectrum visualisation using a microcontroller and MAX7219 displays.

- **Audio-Reactive Spectrum Visualiser: Original Project Proposal**
  Background to the concept, earlier prototypes, proposed extensions and links to relevant signal-processing theory. Provided through Canvas.

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- **Lab-in-a-Box CDIO Development Report**
  Analysis of system behaviour and examples of engineering challenges based on sampling limitations, aliasing, filtering and system validation. Provided through Canvas.

You may also consult relevant component documentation, library documentation, textbooks, lecture materials, technical forums and other reliable online sources.

Learning through experimentation

You are not expected to know the complete solution at the beginning. The purpose of the challenge is to develop your understanding through investigation, experimentation, testing and reflection.

Unexpected behaviour should not automatically be hidden or removed. It may reveal an important limitation, an incorrect assumption or a useful opportunity for improvement.

A partially completed system that has been tested carefully and is well understood may provide stronger evidence of engineering competence than a polished system assembled without understanding.

Capturing your progress

At the end of each session, you will complete a short individual reflection on your contribution, decisions, learning and next steps.

Retain (and upload to Canvas) useful evidence of the project's development, which might include:

- photographs or short videos;
- sketches or system diagrams;
- test results and observations;
- calculations;
- code versions;
- evidence of unsuccessful approaches;
- feedback received; and
- brief explanations of important decisions.

These reflections and artefacts will contribute to your final professional skills portfolio. They should show how your thinking and engineering judgement developed, not merely document a successful final result.

Final demonstration

At the end of the challenge, your team should be prepared to:

- demonstrate the visualiser using appropriate input signals;
- explain how the principal stages of the system work together;
- show evidence of how its performance was tested;
- describe the useful operating range and any important limitations;
- explain one significant decision or improvement;
- identify a possible failure mode or misleading behaviour;
- discuss relevant usability, inclusion, ethical, security or sustainability considerations; and
- identify what you would investigate or improve next.

The emphasis is on the quality of your engineering process, evidence, understanding and reflection, as well as the final prototype.

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